

Phantom Orbit Shadowing for the Collatz Conjecture: Verified Core, Two Barriers, and a Repositioning (v4 — consolidation and limitations)

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Abstract

This is a *consolidation and repositioning* version of an ongoing, AI-assisted personal research program on the Collatz conjecture. It does **not** report progress toward a proof of the conjecture. Its purpose is the opposite and, we argue, equally worthwhile: to state precisely what the program has *rigorously established*, to identify exactly *where* and *why* the program stops, and to record an honest decision about what to do next.

We first recall the verified core, all mechanically checked in Lean 4 with Mathlib and **sorry-free**: an exact congruential shadowing lemma, its no-infinite-(periodic-)shadowing corollary, a family of finite Collatz–Wielandt spectral-radius certificates, a finite phantom-set taxonomy, and an elementary, formally verified bridge reducing the conjecture to a uniform strict-descent hypothesis.

We then analyze the two analytic branches developed after v3 — a weak/operator (averaged) branch and a pointwise first-barrier branch — and reach a definite conclusion about each. The weak branch, even if completed, yields an averaged (“almost-all”) statement and is blocked by the distributional-to-pointwise barrier. For the pointwise branch we show that, *within the present phantom/2-adic formalism*, the desired 2-adic rigidity on the limit point ξ_W of an aperiodic expanding word is *logically equivalent* to the nonexistence of divergent integer orbits, hence cannot act as an independent lever. The reason is a precise *non-transport lemma* (conditional on the standard Lagarias parity-vector coding): the Syracuse parity-vector conjugacy carries eventual periodicity to dynamical pre-periodicity, *not* to rationality. As a by-product we record one new elementary fact, already *Lean-checked* in post-prefix congruence form: no positive integer lies on an expanding cycle (Proposition 11), generalizing the no-infinite-shadowing corollary.

Finally we separate the Collatz problem into its two genuinely distinct hard halves — absence of divergent orbits (barrier: measure-to-pointwise) and absence of nontrivial cycles (barrier: Diophantine approximation of $\log_2 3$) — locate every part of this program relative to those two barriers, and state the resulting repositioning: stop pushing the weak/operator and first-barrier branches as direct routes to the conjecture; redirect effort to the cycle half (where partial results and tools genuinely bite) and to Lean formalization, while keeping the methodology study of AI-assisted mathematics as an explicit goal. All claims here are honest about scope; we make no claim of subcriticality, no spectral gap, and no proof of the conjecture.

Research program note. This is the fourth versioned artifact of a personal program by the author, an independent researcher, exploring whether iterative collaboration with large language model (LLM) assistants can support non-standard attempts at open problems. *Testing the practical limits of such collaboration is an explicit secondary goal, on equal footing with the mathematics.* v4 differs in kind from v1–v3: where earlier versions added constructions and certificates, v4 reports the program reaching — and recognizing — its

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barriers, and documents the resulting strategic decision. We regard the explicit, provable identification of an approach’s own ceiling as a legitimate scientific output. The earlier versions remain available: concept DOI 10.5281/zenodo.20021537; v2 10.5281/zenodo.20098868; v3 record <https://zenodo.org/records/20160154>, DOI 10.5281/zenodo.20160154.

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1 Introduction

The Collatz conjecture asserts that the map $n \mapsto n/2$ (n even), $n \mapsto 3n + 1$ (n odd) sends every positive integer to 1. We work throughout with the accelerated Syracuse map on positive odd integers,

$$S(n) = \frac{3n + 1}{2^{\nu_2(3n+1)}},$$

where ν_2 is the 2-adic valuation; S maps positive odds to positive odds, and Collatz for S is equivalent to the classical statement (Section 3).

What this program built (v1–v3). The program grew from a single empirical observation: “rebel” orbits of S shadow, for finite stretches, periodic rational points of the 2-adic dynamics attached to negative rational fixed points — *phantom cycles* in the terminology of [2], the underlying object being classical [3]. From this, v1–v3 constructed an episode graph on phantom orbits, encoded its critical strongly connected component (SCC) as a finite weighted transfer operator on a refined phase state, proved an exact congruential shadowing lemma, and produced finite Collatz–Wielandt spectral-radius certificates — several of them mechanically verified in Lean 4. v3 [1] framed the result as a *conditional spectral reduction*: if two explicit finite-matrix estimates held (a uniform bound under refinement and a signature-closure condition), the critical SCC would be subcritical.

What changed after v3. Substantial post-v3 work pursued those two estimates along two distinct routes, and this version reports the honest outcome. Neither route reaches the conjecture, and — crucially — we can now say *why*, in each case precisely. The contributions of v4 are:

1. A clean restatement of the verified core (Section 2) and of the elementary descent bridge (Section 3), including the observation that the bridge *reorganizes* the difficulty rather than reducing it.
2. A status report on the two analytic branches (Section 4): a weak/operator (A0) branch, whose finite skeleton is partly formalized but whose natural conclusion is an averaged statement; and a pointwise first-barrier branch, which targets the correct (every- n) statement.
3. The main new result, a *negative* resolution of the currently available phantom/2-adic rigidity route (Section 5): a non-transport lemma (conditional on the standard Lagarias coding) showing that the sought aperiodic rigidity is logically equivalent to the divergence half of the conjecture itself.
4. A small new fact (Proposition 11), Lean-checked in post-prefix congruence form: no positive integer lies on an expanding cycle.
5. A precise two-barrier map of the whole problem (Section 6) and the resulting repositioning (Section 7).

Stance. We make no claim of progress toward Collatz. We claim a verified core, a provable account of where two natural approaches stop, and a defensible plan. Throughout, “rigorous” means proved (and, where stated, machine checked); “empirical” means numerically supported only; “conditional” means valid under a named, unproved hypothesis.

2 The verified core

All statements in this section are formalized in Lean 4 with Mathlib and are `sorry`-, `admit`- and `axiom`-free in the project sources.

2.1 Exact congruential shadowing

For a phantom word w (a nonempty list of positive exponents) with period sum A and length L , let $q_w \in \mathbb{Q}$ be its rational fixed point and ξ_m its 2-adic prefix data.

Lemma 1 (Exact shadowing; `exact_shadowing`). *A positive integer n realizes the first m exponents of w if and only if $n \equiv q_w \pmod{2^{B_m+1}}$, where B_m is the corresponding partial period sum. In particular, b -fold shadowing is governed by the single residue condition $n \equiv q_w \pmod{2^{bA+1}}$.*

This is the iterated step-by-step form of the per-block 2-adic repulsion identity of [2, Prop. 7.4], equivalent in content to the classical inverse formula for parity vectors [4]. We emphasize this equivalence: the 2-adic/phantom encoding is a faithful *re-encoding* of the parity-vector picture, not new information — a point that becomes decisive in Section 5.

Corollary 2 (No infinite periodic shadowing; `no_infinite_period_congruence_expansive`). *For an expanding phantom w (i.e. $3^L > 2^A$), no positive integer shadows w^∞ for all m : the limiting congruence forces $n = q_w$, and $q_w < 0$.*

2.2 Finite spectral-radius certificates

For each truncation depth T and high-bit lift count j , the program builds a finite matrix $\text{FULL}_{T,j} = \text{CORE}_{T,j} + \text{TAIL}_{T,j}$ encoding first-return behavior at the critical node, and applies a weighted Collatz–Wielandt bound [5]. Two certificates are exposed to Mathlib’s real `spectralRadius`:

Theorem 3 (`t10j32HighBitTailSpectralRadiusBound_97_2000`). $\rho(\text{FULL}_{10,32}) \leq 97/2000 = 0.0485$, with no *axiom*, *sorry*, or *admit* in the generated certificate.

Theorem 4 (`k16s16KDeterministicGeneratedSpectralRadiusBound`). *The deterministic finite-residue-cell retention matrix on the compressed (K, b) macro-state space of the $K_0 = 16$ taxonomy SCC (37 states) has exact rational max ratio*

$$\frac{90833233962213}{129559208330288} < \frac{3}{4},$$

with sensitivity confirmed at $\text{lift_bits} \in \{4, 5, 6\}$.

These are exact statements about *declared finite matrices*. They do not assert a spectral gap for any infinite operator, and v4 retracts no part of v3 here: it only stresses the gap between a finite certificate and an infinite conclusion (Sections 4–5).

3 The descent bridge, and why it does not reduce difficulty

Let $S^{[k]}$ denote k -fold iteration of S .

Definition 5 (`UniformStrictDescentHypothesis`). For every positive odd $n \neq 1$ there exists k with $S^{[k]}(n)$ positive, odd, and $S^{[k]}(n) < n$.

Theorem 6 (Lean-verified bridge; `classicalCollatz_of_uniformStrictDescent_proved Bridge`). *Definition 5 implies the accelerated Collatz statement (by strong induction), and the accelerated statement implies the classical Collatz conjecture (by expanding each S -step into one $3n + 1$ step followed by the exact halvings). Both implications are proved in Lean, sorry-free.*

Remark 7 (The bridge reorganizes; it does not reduce). Definition 5 is the classical “every n eventually descends below itself,” which is equivalent to the convergence half of Collatz. Theorem 6 is therefore a faithful, mechanically verified *restatement*, valuable for organizing a formal proof, but it carries the full difficulty. Every elaboration of the bridge in our sources (a global descent cover, finite-model soundness interfaces, branch/loss audits) shares this property: each requires one to *produce* actual strict-descent witnesses for every source class. Such scaffolding isolates the obligation cleanly; it does not lighten it. We state this plainly to forestall the most common misreading of programs of this type.

4 The two analytic branches after v3

The two estimates left open by v3 were attacked along two routes. We describe each and its honest status.

4.1 Branch A: the weak/operator (A0) route

Branch A treats the generated matrices as an averaged row-source kernel K_N on a finite phase alphabet, with source weight μ_N , and studies the weighted row- L^1 prefix drift

$$D_N = \sum_i \mu_N(i) \sum_j |K_{2N}(i, j) - K_N(i, j)|.$$

The target is an A0 weak approximation theorem with named hypotheses A0W1–A0W7 (finite kernel identification; $D_N \rightarrow 0$; an exceptional high- ν_2 source tail; tail/killing control; uniform projection and inclusion bounds for a Banach pair $B_s \rightarrow B_w$; a limit operator convention). Genuine, partly formalized, progress was made:

- **High- ν_2 tail mass, exactly, in Lean** (`TailCount.dyadic_tail_count_mul_le`, `dyadic_tail_mass_le`): for the two-prefix source model, $\mu_N(\nu_2 \geq R) \leq 2^{-R}$. This closes one half of the D_N split rigorously.
- **A bit-length boundary mechanism, in Lean** (`biAffineDelta_dyadicWeight_period_boundary_le` and the `BitLength` development): under a period shift, the discrepancy of the dyadic return weights $2^{-\delta}$ is controlled by the weighted mass of an explicit set of dyadic boundary crossings of cardinality $O(c \log N)$.
- **Exact 2-adic periodicity of bounded words** (script-level, finite): the pure bounded valuation word is eventually periodic modulo 2^M , so the bounded congruential part of D_N cancels *exactly* (top dyadic Haar coefficient 0).
- The finite algebraic bridge (`weighted_action_diff_le`): weighted row- L^1 drift controls the finite $\ell^\infty \rightarrow L^1(\mu_N)$ action error.

Remark 8 (Why Branch A cannot reach Collatz). Every object of Branch A is an *averaged* quantity: $L^1(\mu_N)$, weighted row means, density-type boundary estimates ($O(\log N/N)$). The uniform / supremum analogues do not contract (the diagnostics show p_{95} and maxima persisting; the natural $\mathbb{Z}_2$ martingale Banach space fails). A completed A0 theorem would establish that the finite matrices have a legitimate *weak* operator limit; by the very nature of an averaged norm, the dynamical consequence is an “almost-all” / density statement, not control of every orbit. This is the distributional-to-pointwise barrier that [2, Sect. 9] identifies as the irreducible obstruction to all standard routes. We do not invoke Hennion or Keller–Liverani: the prerequisites (Banach pair, boundedness, compactness, a Lasota–Yorke inequality) are absent. Branch A is therefore retained, if at all, as a *separate* operator-approximation program, not as a route to the conjecture.

4.2 Branch B: the pointwise first-barrier route

Branch B is genuinely different in kind: it is pointwise and parametric, not averaged. On the source cylinder $n = 103 + 256t$, a parametric common prefix $[1, 1, 2, 1, 1, 1]$ (a congruence valid for all t , Lean: `a0SemanticDropCommonPrefix_matches_source_cylinder`) carries the orbit to the endpoint coordinate $593 + 1458t$. One then asks, for every t , whether some matched suffix from the endpoint drops below the source. The machinery built and verified is the *easy direction* plus its arithmetic plumbing:

- if a matched suffix crosses the slope barrier $1458 \cdot 3^{\text{len}} \leq 256 \cdot 2^{\text{sum}}$ and a threshold inequality holds, then t exits (`a0EndpointSuffixExit_of_suffix_threshold_contracting`);

- the threshold is monotone in t (verify at one t_0 , then extend), and matching *forces* a 2-adic congruence on t (`a0EndpointSuffixCongruence_of_matches`).

The descent of the whole cylinder then follows — but only *conditionally*:

Theorem 9 (Conditional cylinder descent; `A0EndpointSuffixExitAll_of_firstBarrier`). *If `A0FirstBarrierExistsAll` (every t reaches a first barrier-crossing suffix) and `A0FirstBarrierThresholdAutomatic` (the threshold holds at that crossing), then every t exits, and every $n = 103 + 256t$ has a strict descent.*

Both hypotheses are `def ... : Prop` in the sources — named, *not* discharged. `A0FirstBarrierExistsAll` says that every orbit on the cylinder eventually contracts past the slope barrier, i.e. that no orbit is eternally expanding. This is precisely the correct (every- t) target. The question of `v4` is whether the first-barrier structure supplies a new lever on it.

5 Negative resolution of the present phantom/2-adic rigidity route

We show that the *currently available* phantom/2-adic route does not supply a new lever, and we prove *why*. We flag the rigor status explicitly: the expanding-cycle fact (Proposition 11) is Lean-checked in congruence form, whereas the non-transport lemma (Lemma 12) and the gate resolution (Proposition 14) are paper-level arguments *conditional on the standard Lagarias parity-vector coding*, not formalized.

Definition 10 (Conjugacy and limit point). Let $\tilde{S} : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ be the continuous extension of S , and let $Q : \mathbb{Z}_2 \rightarrow \prod_{k \geq 1} \{a : a \geq 1\}$, $Q(x)_k = \nu_2(3\tilde{S}^{k-1}(x) + 1)$, be the parity-vector conjugacy (a topological conjugacy of $\tilde{S}$ with the shift). For an infinite word W with all entries ≥ 1 , let $\xi_W = \lim_m \xi_m \in \mathbb{Z}_2$ be the limit of the prefix representatives. If a positive integer n has valuation word W , then $n = \xi_W$.

Proposition 11 (Expanding cycles exclude integers). *No positive integer lies on a periodic orbit of S whose period word v is expanding ($3^{|v|} > 2^{A_v}$). In particular no positive integer has an eventually periodic, eventually expanding valuation word.*

Proof. A periodic point with word v equals the fixed point $q_v = C_v / (2^{A_v} - 3^{|v|})$, with $C_v > 0$ and, in the expanding case, $2^{A_v} - 3^{|v|} < 0$, so $q_v < 0$. Every iterate of a positive odd integer under S is a positive integer (`syracuse_pos`). A positive orbit cannot contain $q_v < 0$. The eventual case follows by applying this to the tail $\tilde{S}^{|u|}(n)$ after a finite prefix u . $\square$

This is the exact extent of what the phantom/2-adic sign argument gives: it generalizes Corollary 2 from the periodic to the eventually periodic obstruction. Its post-prefix congruence form is *Lean-checked*, *sorry-free*, as `no_positive_endpoint_eventually_periodic_expansive_congruence` in `NoInfinite.lean`: after any finite prefix, the positive endpoint of an integer orbit cannot remain, for all numbers of periods, in the congruence classes of an expanding phantom period. It deliberately addresses only the eventually periodic case and does *not* rule out an aperiodic word shadowing different phantom neighborhoods in succession — exactly the gap analyzed next.

Lemma 12 (Non-transport of the rational/periodic dichotomy; standard Lagarias coding). *Under Q , eventual periodicity of W corresponds to pre-periodicity of ξ_W under $\tilde{S}$ (the orbit enters a cycle), not to rationality of ξ_W . Consequently the implication*

$$[W \text{ not eventually periodic} \implies \xi_W \notin \mathbb{Z}_{>0}]$$

is logically equivalent to the nonexistence of divergent positive-integer S -orbits.

Proof. Q is a topological conjugacy, so W is eventually periodic iff ξ_W is pre-periodic under $\tilde{S}$. The contrapositive of the displayed implication reads $[\xi_W = n \in \mathbb{Z}_{>0} \Rightarrow W \text{ eventually periodic}]$, i.e. $[n \in \mathbb{Z}_{>0} \Rightarrow n \text{ is pre-periodic}]$, i.e. every integer orbit eventually enters a cycle — the negation of the existence of a divergent integer orbit. $\square$

Remark 13 (Contrast with binary digits). For the binary-digit map d , “aperiodic digits $\Rightarrow$ irrational” is a free theorem, because d transports eventual periodicity to *rationality*. The valuation/parity vector is a different coordinate system: Q transports eventual periodicity to *pre-periodicity*, and “integer $\Rightarrow$ pre-periodic” is exactly the open divergence problem. This is the precise structural reason no Diophantine rigidity on ξ_W is available in the aperiodic case.

Proposition 14 (Gate resolution: no new lever within the present formalism). *On its hard (aperiodic) part, `AOfirstBarrierExistsAll` carries the full classical difficulty of the divergence half of Collatz, with no additional 2-adic leverage supplied by the currently available phantom/first-barrier formalism. Concretely:*

- (i) *The eventually periodic expanding case is excluded by the sign argument (Proposition 11).*
- (ii) *The aperiodic case admits no rationality / irrationality obstruction on ξ_W (Lemma 12).*
- (iii) *Orbit-wide integrality is automatic (`syracuse_pos/syracuse_odd`) and adds no constraint.*
- (iv) *The only genuinely new datum of the formalism — the sign of the periodic fixed point — is vacuous on both open cases: contractive cycles have $q_w > 0$ (no sign contradiction; the cycle problem is untouched), and aperiodic words have no fixed-point equation at all (the divergence problem is untouched).*

Remark 15 (Consistency with the historical record). This is not surprising; it is the precise articulation, in the program’s own ξ_W language, of why the problem has resisted since the 1970s. Density-one stopping-time results [7, 8] are reachable because the separating tool is statistical *drift* (the mean valuation $2 > \log_2 3$); “every n ” is out of reach because drift does not constrain an individual ξ_W . Both branches of this program hit *this same wall*: Branch A as the distributional-to-pointwise barrier, Branch B as the absence of rigidity on ξ_W — two faces of “density one yes, every n no.”

6 The two barriers, precisely

It is useful to separate the conjecture into its two genuinely distinct hard halves, and to locate every part of this program relative to each.

Half 1: no nontrivial cycles. A cycle is a positive-integer solution of $n = C_w/(2^A - 3^L)$; for $n > 0$ one needs $2^A > 3^L$ (contractive). Ruling out all such reduces to lower bounds on $|A \log 2 - L \log 3|$ for all (A, L) — the Diophantine approximation of $\log_2 3$. Baker-type bounds on linear forms in logarithms give partial exclusions: no 1-cycles [9], no 2-cycles [10], no m -cycles up to $m \approx 68$ [11]; combined with computational verification to $\approx 2^{69}$ [12], any nontrivial cycle must be astronomically large. The full statement is *open*: the available bounds leave an uncovered range, and closing it would need a major advance in the irrationality measure of $\log_2 3$ or a new structural idea. This is a real but *different* wall from Half 2.

Half 2: no divergent orbits. This is the measure-to-pointwise barrier of Section 5/Remark 8: density-one descent is known [7, 6]; every- n descent is not, and no pointwise obstruction is known. This program’s phantom/spectral/first-barrier machinery lives entirely here.

Locating the program.

component	half	status
shadowing lemma + Lean core	both (structure)	rigorous
finite CW certificates	— (finite)	rigorous, finite only
expanding-cycle exclusion (Prop. 11)	Half 1	rigorous (narrow)
weak/operator branch A	Half 2	blocked (averaged)
first-barrier branch B	Half 2	blocked (Prop. 14)

The program contributes a verified *structural* core to both halves and one narrow rigorous exclusion in Half 1, but its analytic effort is concentrated entirely in Half 2, against the harder of the two walls.

7 Decision and planned direction

We record the strategic decision reached, with reasons.

1. **Stop** treating Branch A (weak/operator) and Branch B (first-barrier) as routes to the conjecture. By Remark 8 and Proposition 14, further finite refinement (more T , more j , more cylinders, more suffixes) cannot cross the Half-2 wall *by this route*; it only re-confirms the reformulation. Branch A may continue *only* as a separate, honestly labeled operator-approximation study.
2. **Redirect** mathematical effort to Half 1 (cycles), the only half where partial theorems exist and tools bite. The realistic, finishable target for this program is the *structural* layer (the cycle equation, the Diophantine condition $A/L \approx \log_2 3$, elementary exclusions, the link to computational verification), *formalized in Lean*. We are explicit about the limit: a usable form of Baker’s theorem on linear forms in logarithms is, to our knowledge, not in Mathlib, so the deep analytic exclusions are not cheaply formalizable; the structural layer is.
3. **Consolidate** the verified assets (this paper) so the program’s real output — a **sorry-free** core, finite certificates, a taxonomy, and a precise barrier map — is on the record independently of the conjecture.
4. **Continue** the methodology study (Section 8) as an explicit goal: this phase is itself a data point on how far cross-AI collaboration carries a non-standard attempt, including its capacity to recognize and prove its own ceiling.

What a genuine new idea would have to supply. For Half 2, a single-orbit (non-measure) control of the valuation average — outside the phantom/2-adic circle, since Lemma 12 shows that circle is closed. For Half 1, an improvement in the effective irrationality measure of $\log_2 3$ beyond current Baker-type bounds, or a structural substitute. We claim neither.

8 Methodology: AI-assisted attempt on an open problem

The program is conducted by the author in collaboration with LLM assistants used in a cross-checking configuration (independent assistants proposing and auditing each other’s constructions, with all formal claims funneled through Lean). v1–v3 demonstrated that this configuration can produce nontrivial, machine-verified mathematics (the shadowing core and the finite certificates) and can manage a large, multi-month formalization.

v4 reports a different and, we think, more informative methodological outcome: in this phase the cross-AI process *converged on a precise negative result about its own approach* — the two-barrier map and the gate resolution of Section 5 — rather than on new progress. We regard the ability of such a collaboration to (i) avoid overclaiming, (ii) localize its own obstruction to a single named hypothesis, and (iii) prove that hypothesis equivalent to the open problem, as a positive methodological finding, distinct from progress on the mathematics. The session logs and the four updated working documents (TODO.md, INVENTORY.md, phase10_repro_manifest.md, phase10_master_report.md) record the decision trail.

9 Honest scope and conclusion

Established (rigorous, mechanically verified). The shadowing lemma and its no-infinite-periodic-shadowing corollary; the finite Collatz–Wielandt certificates exposed to Mathlib’s `spectralRadius`; the elementary descent bridge; the verified-core `sorry`-freeness; and, new in this version, the post-prefix congruence form of Proposition 11 (`no_positive_endpoint_eventually_periodic_expansive_congruence`).

Established (on paper, this version; conditional on the standard Lagarias coding). Lemma 12 (non-transport) and Proposition 14: within the currently available phantom/2-adic formalism, the first-barrier route supplies no new lever on Half 2. These are not Lean-formalized and are stated modulo the standard parity-vector conjugacy.

Empirical (not proved). All numerical spectral bounds and decay trends of v1–v3; the finite D_N decay of Branch A.

Missing (and, we argue, behind known barriers). A proof of either half of Collatz. Half 2 is blocked by the measure-to-pointwise barrier, into which both analytic branches of this program provably fall. Half 1 is open behind the Diophantine wall, with partial results but no full exclusion.

Honest claim. We have built and verified a structural core around the Collatz problem, we have determined precisely where two natural analytic approaches stop and proved why, and we have repositioned the program accordingly. We do not claim a proof of the conjecture, nor progress toward one; we claim a clear, verified, and honestly bounded map of an attempt.

Reproducibility

The Lean 4 development, the enumeration and diagnostic scripts, and the working notes referenced here are provided in the supplementary archive. Theorem-to-file correspondences are given inline by Lean declaration name. Earlier versions: concept DOI 10.5281/zenodo.20021537; v3 record <https://zenodo.org/records/20160154>.

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